

Global Sustainability Analysis

Identifying where investments in renewable energy, health, and education have the highest impact.

3 KEY QUESTIONS

01

Where would renewable energy investments have the greatest impact?

02

Which countries show the most critical social indicators?

03

Strong economy, but energy inefficient?

Dataset & Approach

global_country

195 countries · 35 features

Demographics, GDP, CO₂,
fertility rate, health & education
indicators, land area

JOIN

sustainable_energy

~2,900 rows · 20 features

2000–2019 · Renewable energy %,
energy intensity, electricity access,
GDP per capita, CO₂ emissions

ANALYSIS PIPELINE

1

Table creation & linking via
FK on country name

2

CTEs for multi-step filtering
(2017–2019 data window)

3

Subquery benchmarks
against global averages

4

Multi-criteria ORDER BY for
impact prioritization

Where would renewable energy investments have the greatest impact?



Low renewable share

High reliance on fossil fuels → largest transformation potential



Limited electricity access

Below world average → infrastructure urgently needed



High CO₂ emissions

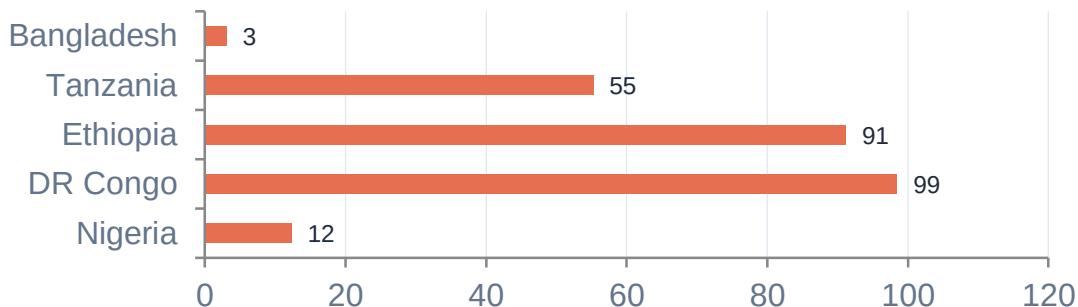
Clean energy here = maximum pollution reduction



Large / dense population

Maximize reach: more people impacted per investment

Top Priority Countries (sorted by CO₂, then lowest renewable %)



Key Insight

Countries like Nigeria and Bangladesh combine high CO₂ emissions, large populations, and very low renewable energy share — making them prime candidates for high-impact energy investments.

Which countries show the most critical social indicators?



Life expectancy

Below world average
→ urgent health need



Tertiary education

Below world average
→ low human capital



Electricity access

Below world average
→ limits all interventions



High fertility rate

Above world average
→ structural challenge

TOP 5 COUNTRIES REQUIRING IMMEDIATE SUPPORT

Country	Life Exp.	Tertiary Edu.	Electricity %	Fertility Rate
Chad	52.5	2.9%	8.4%	5.79
Central Afr. Rep.	53.1	2.1%	14.5%	5.54
Niger	62.4	2.4%	18.7%	6.82
Mali	59.3	5.2%	35.2%	5.95
Burkina Faso	61.2	4.8%	43.5%	5.21

Countries with strong economic potential but energy inefficiencies

△ □ UNTAPPED POTENTIAL

Singapore

GDP growth +112% · Renewables 3.2%

High GDP growth, near-zero renewables

United States

GDP growth +45% · Renewables 11.4%

Massive economy, large fossil dependency

Malta

GDP growth +87% · Renewables 7.8%

Small, dense — ideal for pilot projects

✓ MODEL COUNTRIES

Uruguay

GDP growth +68% · Renewables 62.3% · Tertiary edu 0.63

Consistent renewable policy since 2008

Lithuania

GDP growth +55% · Renewables 41.1% · Tertiary edu 0.79

Strong education + energy transition

Iceland

GDP growth +33% · Renewables 99.8% · Tertiary edu 0.82

Nearly 100% renewable electricity

💡 Model countries show above-average tertiary education enrollment (global avg ~0.37), reinforcing the link between education and renewable adoption.

Conclusions & Next Steps

Q1 — Renewable Energy

Nigeria, Bangladesh and DR Congo offer the highest transformation potential: large populations, high CO₂, and very low renewable share. Targeted infrastructure investment here maximizes environmental ROI.

Q2 — Social Indicators

Chad, Niger, and Mali require immediate multi-sector support. Low electricity access compounds the effects of poor health and education outcomes, creating a reinforcing cycle of underdevelopment.

Q3 — Economic Potential

Singapore and Malta are ideal pilot project candidates: strong GDP growth, existing infrastructure, but low renewable share. Lessons can be scaled to larger economies like the US.